

ST 538 Group 4 Project 2

Introduction

Vehicle loan default prediction presents a distinct set of challenges relative to other retail credit products. Unlike mortgage or personal loan portfolios, two-wheeler financing in emerging markets such as India serves a large population of first-time borrowers with little or no bureau history, making traditional score-based underwriting insufficient on its own. This analysis uses the CRIF High Mark vehicle loan dataset to develop a feature set capable of identifying first-EMI default risk across both scored and thin-file borrower segments.

Data and Processing

The dataset comprises 233,154 disbursed loans with a binary default indicator capturing payment failure on the first EMI due date. A notable characteristic of this population is that 56% of borrowers carry no scoreable bureau history, which necessitates treating thin-file and scored segments with care rather than imputing or ignoring the distinction. Among scored borrowers, the CRIF CNS score serves as the single strongest individual predictor and is encoded here as a 20-level ordered factor preserving the full risk ranking across bands.

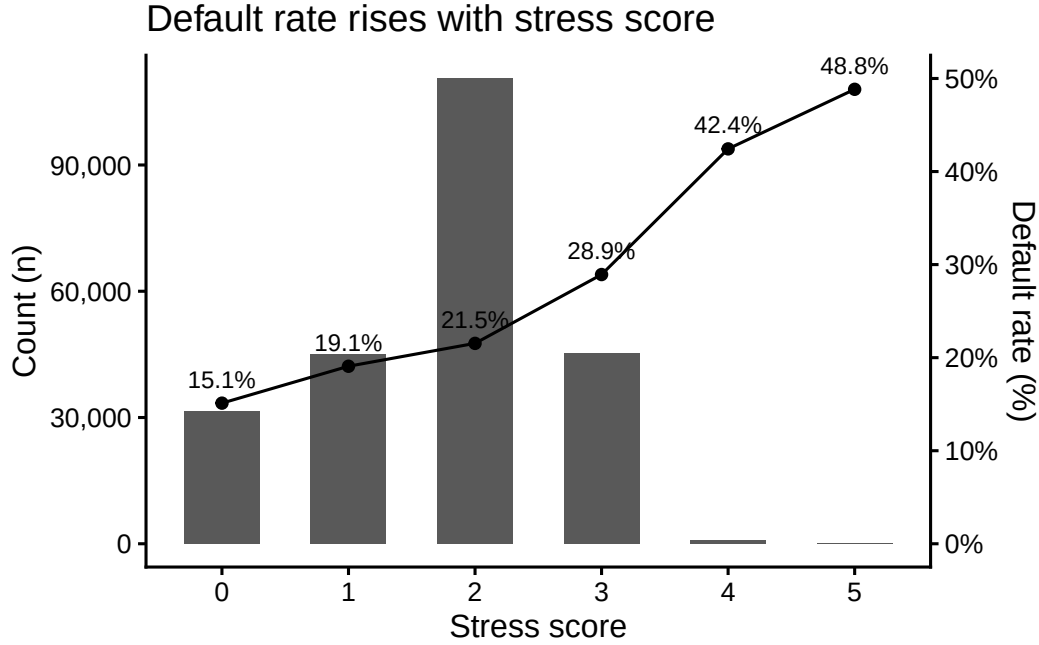
Feature engineering focuses on translating domain knowledge into model-ready variables: loan structure risk (LTV, implied down payment), borrower demographics (age at disbursement, employment type), identity verification depth (KYC document count), credit utilisation and delinquency history across primary and secondary accounts, EMI burden relative to existing obligations, and recent bureau activity including new account uptake and hard enquiries.

A composite stress score aggregating six binary risk flags is also constructed as a diagnostic and interpretability aid. Initial cross-tabulation confirms that the stress score tracks default rate monotonically, rising from 15% at score 0 to 49% at score 5, providing early evidence that the engineered features carry meaningful signal.

Summary of EDA and feature engineering:

Table 1: Default Rate by Stress Score

Stress score	Total (n)	No default	Default	Default rate
0	31,573	26,802	4,771	15.1%
1	44,945	36,371	8,574	19.1%
2	110,556	86,748	23,808	21.5%
3	45,181	32,110	13,071	28.9%
4	813	468	345	42.4%
5	86	44	42	48.8%



Thirty-five percent of loans are high loan to value (LTV) of >80%, combined with a large portion of borrowers having thin credit history (%), it shows a meaningful concentration of risk. The vast majority (94%) of borrowers fall in the prime age band, 13,812 young borrowers (<22 years of age) are worth watching. Fifty-six percent of borrowers are “thin file” meaning they have <12 month of credit history. This makes sense, in India about half of first time car buyers are using lending for the first time. Thin and normal are two different populations, we may want to split the dataset and create a model for thin credit and one for normal, or focus our efforts on normal only—the model will likely suffer trying to work on both.

Stress score is working well as a risk signal. Default raises from 15% at score 0 to ~49% at score 5 (3x). Most of the population sits at 2 (21% default), probably thin-file + high-LTV, but no active delinquency. The jump from 2 to 3 is steep (21-29%), `pri_any_overdue_flag` and/or `recent_delinquency_flag` are likely doing a lot of work. Scores 4 and 5 are small segments with high default rates (42-49%), it’s really up to us if we want to keep them, probably should.

Score 0 is not zero (15.1%), even the cleanest borrowers carry default risk. We can use this `stress_score` as a standalone feature, but it's worth noting that components might actually capture more non-linear interactions than the score itself.

Building Classifiers and Results

Table 2: Random Forest Model Comparison

Model	Dataset	TN	FP	FN	TP	Accu- racy	Re- call	Preci- sion	Speci- ficity	F1
rf.sim1	Training	57124	395	14256	355	0.797	0.024	0.473	0.993	0.046
rf.sim1	Valida- tion	12415	94	3056	78	0.799	0.025	0.453	0.992	0.047
rf.sim2	Training	57132	387	14246	365	0.797	0.025	0.485	0.993	0.048
rf.sim2	Valida- tion	12407	102	3058	76	0.798	0.024	0.427	0.992	0.046
rf.final	Training	43542	13977	7518	7093	0.702	0.485	0.337	0.757	0.398
rf.final	Valida- tion	9345	3164	1646	1488	0.693	0.475	0.320	0.747	0.382

After feature extraction and variable selection, we split the dataset into training, validation, and testing subsets using a 70/15/15 split, resulting in approximately 72,000 observations in the training set and 15,000 observations each in the validation and testing sets. We began by fitting a default random forest classifier to the training data and evaluating its performance on both the training and validation datasets. Due to the strong class imbalance in the loan default variable, the initial models showed poor sensitivity for detecting loan defaults, despite relatively high overall accuracy. We then adjusted model parameters and explored additional tuning strategies, including the use of `tuneRF()` to optimize the `mtry` parameter and balanced sampling to improve classification performance. The tuned model substantially improved recall for default predictions, though it also increased the number of false positive classifications. While the final random forest model showed meaningful improvement over the baseline models (Table 2), prediction performance for loan defaults remained moderate, indicating that additional tuning or alternative classification methods may be necessary for highly imbalanced loan data. In future work, we plan to continue tuning the random forest model and compare its performance against at least two additional classification approaches.